

CryptMT3 Stream Cipher^{*}

Makoto Matsumoto¹, Mutsuo Saito², Takuji Nishimura³, and Mariko Hagita⁴

¹ Dept. of Math., Hiroshima University,
`m-mat@math.sci.hiroshima-u.ac.jp`

² Dept. of Math., Hiroshima University,
`saito@math.sci.hiroshima-u.ac.jp`

³ Dept. of Math. Sci., Yamagata University,
`nisimura@sci.kj.yamagata-u.ac.jp`

⁴ Dept. of Info. Sci., Ochanomizu University,
`hagita@is.ocha.ac.jp`

Abstract. CryptMT version 3 (CryptMT3) is a stream cipher obtained by combining a large LFSR and a nonlinear filter with memory using integer multiplication. Its period is proved to be no less than $2^{19937} - 1$, and the 8-bit output sequence is at least 1241-dimensionally equidistributed. It is one of the fastest stream ciphers on a CPU with SIMD operations, such as Intel Core 2 Duo.

1 Introduction

In this article, we discuss pseudorandom number generators (PRNGs) for stream ciphers. We assume that the PRNG is implemented in software, and the platform is a 32-bit CPU with enough memory and fast integer multiplication.

Our proposal [5][6] is to combine a huge state linear generator (called the mother generator) and a filter with memory, as shown in Figure 1.

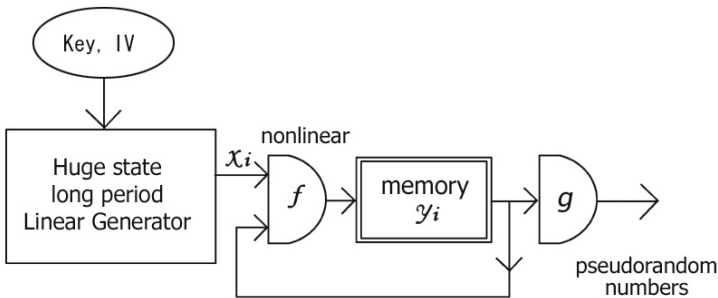


Fig. 1. Combined generator = linear generator + filter with memory

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Definition 1. (*Generator with a filter with memory.*) Let X be a finite set (typically the set of the word-size integers). The mother generator G generates a sequence $x_0, x_1, x_2, \dots \in X$. Let Y be a finite set, which is the set of the possible states of the memory in the filter. We take a $y_0 \in Y$. Let $f : Y \times X \rightarrow Y$ be the state transition function of the memory of the filter, that is, the content y_i of the memory is changed by the recursion

$$y_{i+1} := f(y_i, x_i).$$

The output at the i -th step is given by $g(y_i)$, where $g : Y \rightarrow O$ is the output function which converts the content of the memory to an output symbol in O .

In a previous manuscript [5], we chose the mother generator to be Mersenne Twister [4], which generates a sequence of 32-bit integers by an $\mathbb{F}_2$ -linear recursion. The filter is given by

$$f(y, x) := y \times (x|1) \bmod 2^{32}, \quad g(y) := 8 \text{ MSBs of } y \quad (1)$$

where $(x|1)$ denotes x with LSB set to 1, and 8 MSBs mean the 8 most significant bits of the integer y . Initially, the memory is set to an odd integer y_0 . This is CryptMT version 1 (CryptMT1). There has been no attacks reported to this generator (even non-practical attacks). We introduced CryptMT version 2 [7] and version 3 [8], not to improve the security, but to improve the speed of initialization and generation. This manuscript is based on [8]. Theoretical analysis of this type of generators is developed in [9], where the quasigroup property of the filter plays the role of “balanced filter”.

2 CryptMT3: A New Variant Based on 128-Bit Operations

Modern CPUs often have single-instruction-multiple-data (SIMD) operations. Typically, a quadruple of 32-bit registers is considered as a single 128-bit register. CryptMT3 proposed here is a modification of the CryptMT version 1, so that it fits to the high-speed SIMD operations.

2.1 Notation

Let us fix the notations for 128-bit integers. A bold italic letter $\mathbf{x}$ denotes a 128-bit integer. It is a concatenation of four 32-bit registers, each of which is denoted by $\mathbf{x}[3], \mathbf{x}[2], \mathbf{x}[1], \mathbf{x}[0]$, respectively, from MSB to LSB.

The notation $\mathbf{x}[3][2]$ denotes the 64-bit integer obtained by concatenating the two 32-bit integers $\mathbf{x}[3]$ and $\mathbf{x}[2]$, in this order. Similarly, $\mathbf{x}[0][3][2][1]$ denotes the 128-bit integer obtained by permuting (actually rotating) the four 32-bit integers in $\mathbf{x}$. Thus, for example, $\mathbf{x} = \mathbf{x}[3][2][1][0]$ holds.

An operation on 128-bit registers that is executed for each 32-bit integer is denoted with the subscript 32. For example,

$$\mathbf{x} +_{32} \mathbf{y} := [(\mathbf{x}[3] + \mathbf{y}[3]), (\mathbf{x}[2] + \mathbf{y}[2]), (\mathbf{x}[1] + \mathbf{y}[1]), (\mathbf{x}[0] + \mathbf{y}[0])],$$